

1.1 dokážeme pomocí axiomů skalárního součinu
 $\frac{1}{4}(\|u+v\|^2 - \|u-v\|^2) = \frac{1}{2} \operatorname{Re} \langle u, v \rangle$ ^{za pomoci (SSS), (SL1), (SL2)}

$$\stackrel{(SL1)}{\stackrel{(SL2)}{=}} \frac{1}{4} (\langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle - \langle u, u \rangle - \langle u, v \rangle - \langle v, u \rangle + \langle v, v \rangle) =$$

$$\stackrel{(SSS)}{=} \frac{1}{4} (2\langle u, v \rangle + 2\overline{\langle u, v \rangle}) =$$

$$= \frac{1}{2} (\langle u, v \rangle + \overline{\langle u, v \rangle}) \stackrel{(a+bi) + \overline{(a+bi)} = (a+bi) + (a-bi) = 2a}{=} \frac{1}{2} (2 \operatorname{Re} \langle u, v \rangle) = \operatorname{Re} \langle u, v \rangle$$

$$\stackrel{(SL1)(SL2)}{\downarrow} \frac{1}{4} (\|u-iv\|^2 - \|u+iv\|^2) \stackrel{(a+bi) - \overline{(a+bi)} = (a+bi) - (a-bi) = 2bi}{=} \frac{1}{4} (\langle u, u \rangle - i\langle u, v \rangle + i\langle v, u \rangle + (i(-i))\langle v, v \rangle - (\langle u, u \rangle + i\langle u, v \rangle - i\langle v, u \rangle + (i(-i))\langle v, v \rangle))$$

$$\stackrel{(SSS)}{=} \frac{1}{4} (-2i\langle u, v \rangle + 2i\overline{\langle u, v \rangle}) =$$

$$= -\frac{i}{2} (\langle u, v \rangle - \overline{\langle u, v \rangle}) \stackrel{(a+bi) - \overline{(a+bi)} = (a+bi) - (a-bi) = 2bi}{=} \frac{-i}{2} (\operatorname{Im} \langle u, v \rangle \cdot 2i) = \operatorname{Im} \langle u, v \rangle$$

$$\begin{aligned} (a+bi) - \overline{(a+bi)} &= \\ &= (a+bi) - (a-bi) = 2bi \end{aligned}$$

